

Humphrey Yang

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EDUCATION

Australian National University

Canberra, ACT

Ph.D. in Economics

February 2024 – Now

Bachelor of Applied Data Analytics

July 2019 – June 2023

Bachelor of Politics, Philosophy, and Economics

July 2018 – July 2021

Rewards: Chancellor's Letter of Commendation / Smarterknowledge Prize for Data Wrangling

EMPLOYMENT HISTORY

Research Assistant | RSFAS, ANU

July 2023 – Now

- Assisting Professor Yanrong Yang in conducting research and experiments on the topic of **Benign Overfitting for Complicated Data Inference** under the ARC Discovery Project (DP230102250).
- Conducting large-scale simulations to investigate the model behavior when applied to high-dimensional data using high-performance computing packages.

Research Assistant | QuantEcon

July 2022 – Now

- Contributed to the lectures and workshops on scientific computing and modeling in quantitative economics and specialized lectures using scientific and high-performance computing libraries (i.e., **Google JAX**) in Python.
- Contributed to projects such as [QuantEcon.py](#) library for quantitative economics and finance and application of Dynamic Mode Decomposition in income distribution.

Research Affiliate | Data 61, CSIRO

March 2023 – July 2023

- Conducted research to design attacks to produce responsible generative AI, especially in Large Language Models (LLMs) using **reinforcement learning and adversarial training**.
- Researched and designed reinforcement learning models for LLMs to induce undesirable behaviors from LLMs, and employed appropriate metrics and framework to define action, state, and reward.

Summer Research Internship | Data 61, CSIRO

November 2022 – February 2023

- Researched to explore the landscape of **copyright protection in generative AI** and published a research poster to the **ACM Web Conference**.
- Implemented and measured **GAN and diffusion models** using CSIRO Bracewell HPC and improved their methodologies based on the results.

PUBLICATIONS AND RESEARCH PROJECTS

Published

QuantEcon.py: A community-based Python library for quantitative economics

with Quentin Batista, et al. *Journal of Open Source Software*, 2024

Copyright Protection and Accountability of Generative AI: Attack, watermarking and attribution

with Haonan Zhong, et al. *Companion Proceedings of the ACM Web Conference*, 2023

Working Paper

Benign Overfitting in Linear Regression with Separable Variance-Covariance Structure

September 2023 – Now

Supervisors: Dr Yanrong Yang and Dr Hanlin Shang

Privacy-preserving Record Linkage in the Presence of Missing Data using Graph Learning

April 2023 – Now

Supervisors: Dr Anushka Vidanage and Dr Thilina Ranbaduge

TEACHING

RSFAS, Australian National University

Principles of Mathematical Statistics for Actuarial Studies (STAT6013) – Tutor

Semester 1, 2023

- Graduate course in mathematical statistics covering **Probability, Discrete and Continuous Probability Distributions, Multivariate Probability Distributions, Estimation, and Credibility Theory**.

Statistical Learning (STAT6040) – Tutor

Semester 1, 2023

- Graduate course in mathematical statistics covering research and developments in statistical concepts and high-performance statistical computing on topics including **Fundamental Theories in Statistical Learning and various parametric and non-parametric methods and model evaluation methods**.

Graphical Data Analysis (STAT3011/STAT4026/STAT7026) - Marking Assistant

Semester 2, 2022

- Graduate course in principles of data presentation and statistical graphics covering techniques for constructing **scientific visualizations**.

Quantitative Research Methods (STAT1008) – Tutor

Semester 2, 2022

- Undergraduate course in introductory mathematical statistics and statistical computing in R.

ACADEMIC REFEREES

Professor John Stachurski
Professor, Co-Chair of QuantEcon
Australian National University
Email: john.stachurski@anu.edu.au

AsPr Yanrong Yang
Associate Professor, RSFAS
Australian National University
Email: yanrong.yang@anu.edu.au

Dr Tina Wu
Research Scientist
CSIRO
Email: tina.wu@data61.csiro.au

ADDITIONAL INFORMATION

Technical skills:

- Proficient in Python, SQL, R, PyTorch, Bash, LaTeX, and Markdown.
- Familiar with C, Java, Julia, high-performance computing, and Sphinx.

Languages:

- English (Proficient; IELTS 8.5), Mandarin Chinese (Native).